LLMS; Alignment, Evaluation, System Design and whatnot

Opening the black box..

$$\mathcal{L}_{pretrain} = -\sum_{t} \log P(x_t | x_{<t})$$

What will the base LLM output for this?

What is the capital of France

A: What is its population?

B: Paris

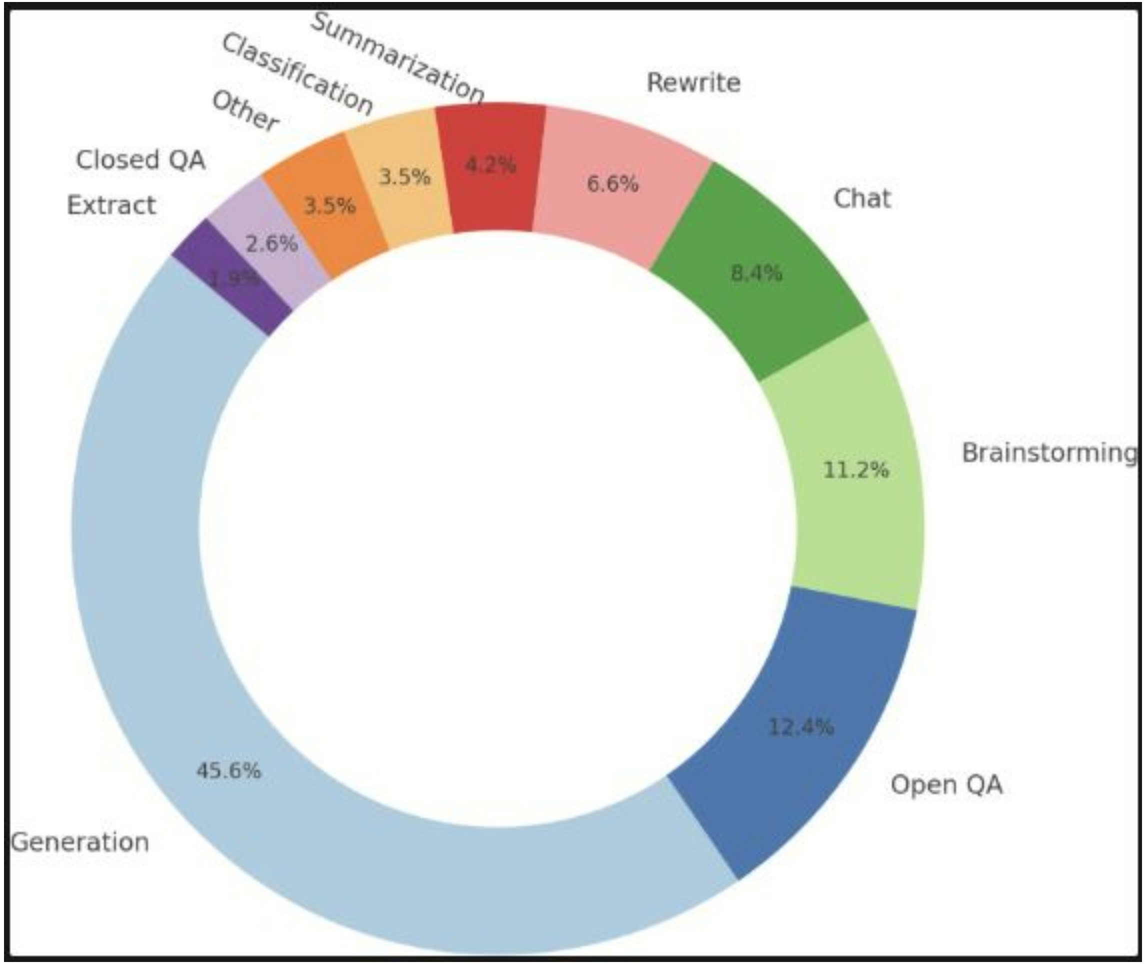
Paradigm Shift

- Pre-training: (Massive text , NTP)
- Post-training (Instruction Tuning, Preference Optimization)
- Base Model = "Document Completer"
- Instruct Model = "Helpful Assistant."

Supervised Fine-tuning (SFT) -> Behavior Cloning

- Dataset -> Model -> SFT Model
- {"messages": [{"role": "user", "content": "..."}, {"role": "assistant", "content": "..."}]}
- Aka Instruction Tuning
- The model isn't learning *why* an answer is good; it's just mimicking the style of the human annotator. I

- What is the correct response?
  - Adding more context to the question: “for a family of six?”
  - Adding follow-up questions: “What ingredients do I need? How much time would it take?”
  - Giving the instructions on how to make pizza. [Behavior cloning]


Summarization
Classification
Other
Closed QA
Extract
Rewrite
Chat
Brainstorming
Open QA
Generation
4.2%
3.5%
3.5%
2.6%
1.9%
6.6%
8.4%
11.2%
12.4%
45.6%

Preference-finetuning

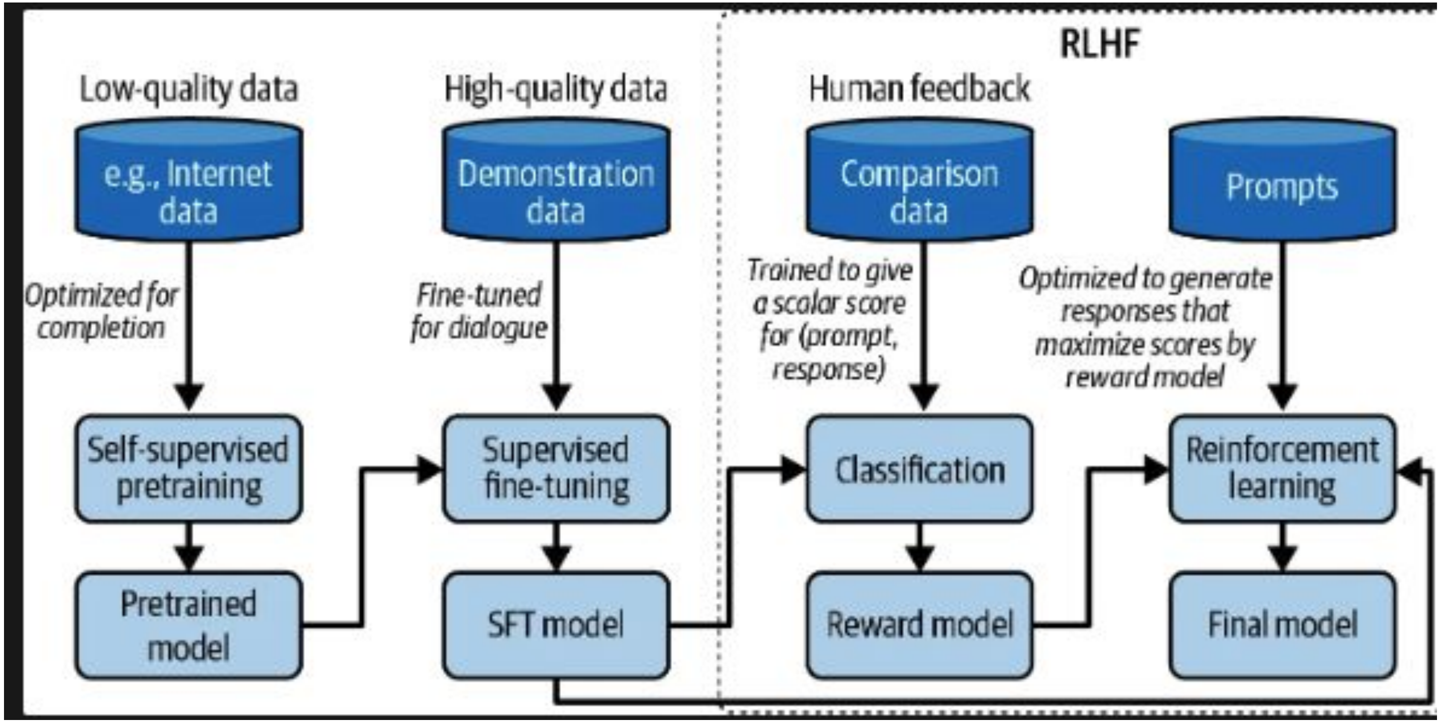
- AI faces ethical challenges in responding to questions about controversial topics like abortion, gun control, the Israel–Palestine conflict, disciplining children, marijuana legality, universal basic income, or immigration.
- Defining and detecting potentially controversial issues is difficult.
- Any response to a controversial issue risks upsetting some users.
- Excessive censorship can make the model boring and drive users away.

Shameless plug

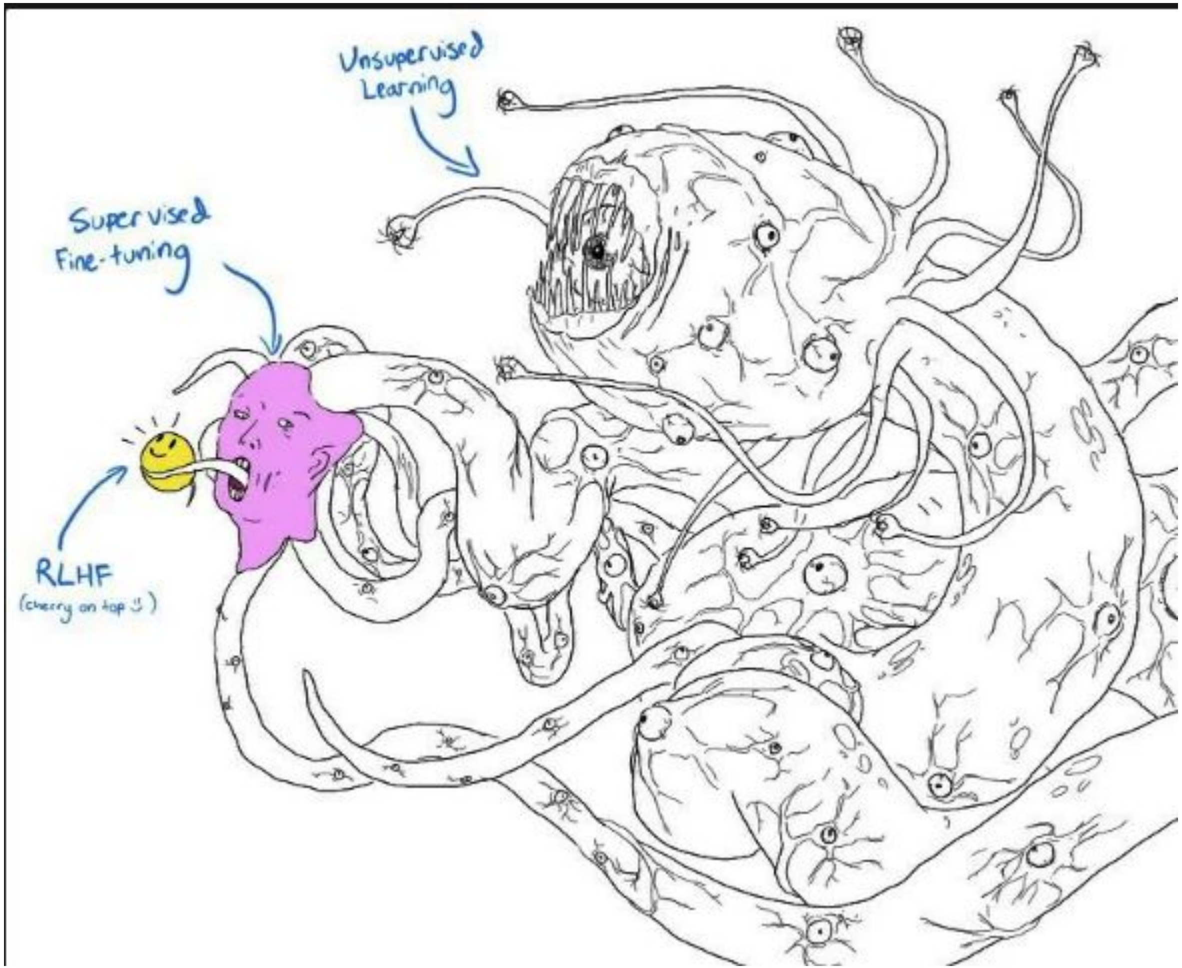
RLHF Pipeline aka why ChatGPT became successful

1. SFT
2. Reward Modelling (Ranking)
3. RL Optimization
   - **PPO (Proximal Policy Optimization).**
   - **Reward Hacking / Specification Gaming.**
   - **KL Divergence Penalty.**

- **Beware of Reward Hacking!**


RLHF
Low-quality data
e.g., Internet data
Optimized for completion
Self-supervised pretraining
Pretrained model
High-quality data
Demonstration data
Fine-tuned for dialogue
Supervised fine-tuning
SFT model
Human feedback
Comparison data
Trained to give a scalar score for (prompt, response)
Classification
Reward model
Prompts
Optimized to generate responses that maximize scores by reward model
Reinforcement learning
Final model

basically….



Unsupervised Learning
Supervised Fine-tuning
RLHF
(cherry on top ツ)

Detour; BLEU, ROGUE

J’ai mangé trois filberts.

I have eaten three hazelnuts.

I ate three filberts.

You came up with this translation:

I ate three hazelnuts.

How can I assign a single numerical score to this translation that tells us how “good” it is using only the provided reference sentences and the neural output?

Knowing the potential best score allows us to calculate the **distance between it and the actual score. This distance provides feedback during training,** indicating if a change moves the score closer to the ideal, and enables comparison of trained systems on the same task.

Unigram precision

- Look at each word in the output sentence.
- Assign a score of 1 if the word appears in any of the reference sentences, and 0 if it doesn't.
- Divide the total number of words that scored 1 by the total number of words in the output sentence.
- This normalizes the count to a value between 0 and 1.

Problem: This is also correct

Three three three three.

Capp the word count?

Ate hazelnuts I three

- Generally, BLEU scores are based on an average of unigram, bigram, trigram and 4-gram precision
- [Ate hazelnuts]

- [hazelnuts I]

- [I three]

-
-

I ate.

- The brevity penalty is a measure that penalizes sentences shorter than the reference translations.
- It is calculated by comparing the output sentence length to the length of the closest reference sentence in length.
- If the output is as long as or longer than any reference sentence, the penalty is 1 (no change to the score).
- If the output is shorter than any reference sentence, a formula is applied: $e^{1 - (\text{Closest Reference Length} / \text{Output Length})}$.
- The closer the shortest reference sentence is to being long, and the shorter the output sentence is, the closer the brevity penalty gets to zero.

Pros

- It’s fast and easy to calculate, especially compared to having human translators rate model output.
- It’s ubiquitous. This makes it easy to compare your model to benchmarks on the same task.

Cons

- It doesn’t consider meaning
- It doesn’t directly consider sentence structure
- It doesn’t handle morphologically rich languages well
- It doesn’t map well to human judgements

Original (French): J’ai mangé la pomme.

Reference translation: I ate the apple.

BLEU ranks these bad

I consumed the apple.

I ate an apple.

I ate the potato.

- Mismatches incur a lower penalty for common n-grams (e.g., "of the") and a higher penalty for rarer n-grams (e.g., "buffalo buffalo").

BLEU doesn’t handle morphologically-rich languages well

ROGUE

ROUGE-N: Measures overlap of N-grams (unigrams, bigrams).

ROUGE-L: Measures the Longest Common Subsequence (LCS). This accounts for sentence structure similarity better than simple N-grams.

- It focuses on recall rather than precision.
- ROUGE measures how many n-grams in the reference translation appear in the output.
- This is the reverse of what BLEU measures.

Et al..

- Perplexity is a measure from information theory, often used for language modeling, that measures how well the learned probability distribution of words matches that of the input text.
- Word error rate (WER) is a common metric in speech recognition that measures the number of substitutions, deletions, and insertions in the output sequence compared to a reference input.
- The subtree metric (STM) compares the parses for the reference and output translations and penalizes outputs with different syntactic structures.
- METEOR is similar to BLEU but includes additional steps like considering synonyms and comparing word stems. Unlike BLEU, it is explicitly designed to compare sentences rather than corpora.
- Translation error rate (TER) measures the number of edits needed to change the original output translation into an acceptable human-level translation.

How to get ‘semantics’...

Evaluation

First, the more intelligent AI models become, the harder it is to evaluate them.

However, for an open-ended task, for a given input, there are so many possible correct responses. It’s impossible to curate a c`omprehensive list of correct outputs to compare against.

Details such as the model architecture, training data, and the training process can reveal a lot about a model’s strengths and weaknesses. Without those details, you can evaluate only a model by observing its outputs

Benchmarks

- story_cloze, MATH, TruthfulQA (single task)
- GLUE (2018) (multi task)
- SuperGLUE (2019)
- NaturalInstructions (2021)
- MMLU (2020), MMLU Pro (2024)
- Previous metrics: cross entropy, perplexity, BPC, and BPB
-

Entropy

- Entropy measures how difficult it is to predict what comes next in a language.
- The lower a language’s entropy, the more predictable that language is (less information carried per token).
- Example: "The cat sat on the ____" illustrates how surprising or informative the next word is.
- Low Entropy = Very predictable.
- High Entropy = Very surprising or random.

Cross entropy

- it is for the language model to predict what comes next in this dataset.
    - Determined by: The training data's predictability
    - How the distribution captured by the language model diverges from the true distribution of the training data
- **P** = the **true distribution** (i.e. the actual probability of how language behaves — from your training data)
- **Q** = the **model's learned distribution** (i.e. what your model *thinks* those probabilities are)
- ideally, you'd like **Q to match P**
- **H(P)** measures the uncertainty in the actual language. Or, $H(P) = - \sum P(x) * \log(P(x))$
- **H(P, Q),** This is the **model's loss**: "How surprised is your model Q when the world behaves like P?" $H(P, Q) = - \sum P(x) * \log(Q(x))$
- $D\_KL(P \| Q) = H(P, Q) - H(P)$

Perplexity is the exponential of entropy and cross entropy. • the number of equally-likely guesses that amount of info represents, Perplexity is the number of equally likely options the model thinks it has to choose from (value 1 ← more confident, gets confused as number is higher)

Functional Correctness (open-ended)

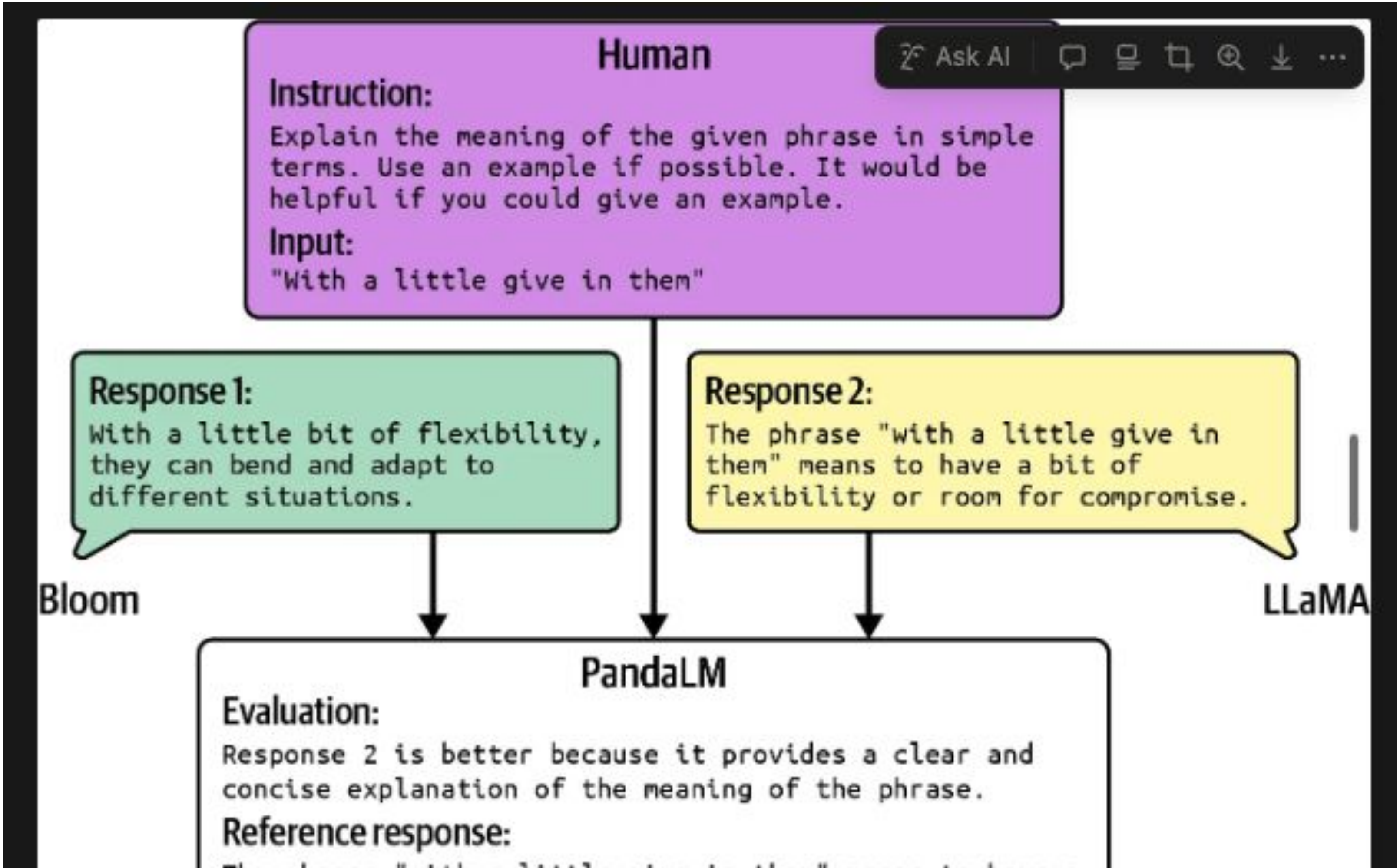
- code generation → *execution accuracy*
- OpenAI's HumanEval , Python

Reference Data

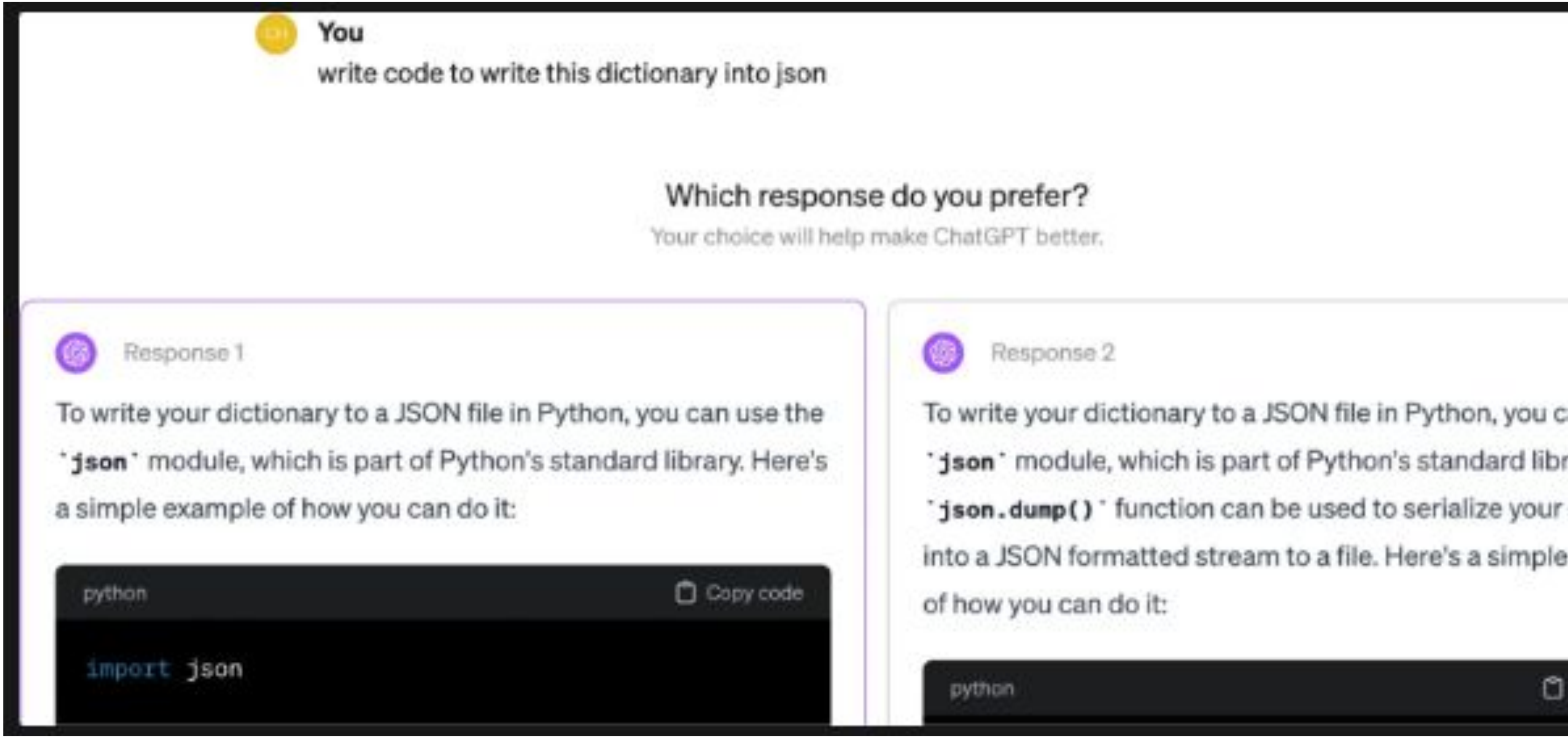
- *ground truths* or *canonical responses*. (French → English, actual translation)
- Simiarlity (exact, lexical, symantic)
- Lexical: BLEU, ROUGE, METEOR++, TER, and CIDE
- Symantic: Vectors (BERTScore, MoverScore)

LLM as a Judge

[LangChain's *State of AI*] *(58% by AI)*


Ask AI
Human
Instruction:
Explain the meaning of the given phrase in simple terms. Use an example if possible. It would be helpful if you could give an example.
Input:
"With a little give in them"
Response 1:
With a little bit of flexibility, they can bend and adapt to different situations.
Response 2:
The phrase "with a little give in them" means to have a bit of flexibility or room for compromise.
Bloom
LLaMA
PandaLM
Evaluation:
Response 2 is better because it provides a clear and concise explanation of the meaning of the phrase.
Reference response:

- **Comparative Evaluation**
    - `Not for everything if answer is Yes/No for example.
    - Asking users to pick can also cause user frustration. Imagine asking the model a math question because you don’t know the answer


You
write code to write this dictionary into json
Which response do you prefer?
Your choice will help make ChatGPT better.
Response 1
To write your dictionary to a JSON file in Python, you can use the `json` module, which is part of Python's standard library. Here's a simple example of how you can do it:
python
Copy code
import json
Response 2
To write your dictionary to a JSON file in Python, you c
`json` module, which is part of Python's standard libr
`json.dump()` function can be used to serialize your
into a JSON formatted stream to a file. Here's a simple
of how you can do it:
python

- Self-Preference Bias: Models tend to rate their own outputs higher than outputs from other models.
- Example: A GPT-4 judge will prefer a GPT-4 response over a Claude response.
- Result (Filter Bubble): This bias creates a "filter bubble" where models are effectively trained to sound like their judge.

models are usually tested domain-wise

exact evaluation

coding $\rightarrow$ functionial correctness efficiency → time taken for the sql query , memory usage opical content → mcqs

Dataset: TruthfulQA

Benchmarking

- Goodhart's Law: "When a measure becomes a target, it ceases to be a good measure." -> they have likely "seen" the test questions during training.
-

Benchmarking

- **A benchmark for competitive math (MATH)**
- **One each for legal (LegalBench), medical (MedQA),**
- **and translation (WMT 2014)Two for reading comprehension—answering questions based on a book or a long story (NarrativeQA**
- **and OpenBookQA)Two for general question answering (Natural Questions under two settings, with and without Wikipedia pages in the input)**
- A benchmark for competitive math (MATH)
- One each for legal (LegalBench), medical (MedQA), and translation (WMT 2014)
- Two for reading comprehension—answering questions based on a book or a long story (NarrativeQA and OpenBookQA)
- Two for general question answering (Natural Questions under two settings, with and without Wikipedia pages in the input)

SelfChheckGPT

- Ask the model the same question 5 times with a high temperature (high randomness).
- Compare the 5 answers.
- Consistency implies Factuality: If the model "knows" the fact (e.g., "Paris is the capital of France"), all 5 answers will be factually consistent, even if worded differently.
- Divergence implies Hallucination: If the model is hallucinating, the 5 answers will likely contradict each other (e.g., giving 5 different dates for an event).